Chapter 5

Binary stars and stellar masses

If we are to understand the workings of stars, it is important to know their masses. The best way to measure the mass of an object is to measure its gravitational influence on another object. (When you stand on a bathroom scale, you are measuring the Earth's gravitational effect on your mass.) For stars, we are fortunate to be able to measure the gravitational effects from pairs of stars, called *binary stars*.

5.1 Binary stars

Many stars we can observe appear to have companions, the two stars orbiting their common center of mass. It appears that approximately half of all stars in our galaxy are in binary systems. By studying the orbits of binary stars, we can measure the gravitational forces that the two stars exert on each other. This allows us to determine the masses of the stars.

We classify binaries according to how the companion star manifests its presence:

Optical double. This is not really a binary star. Two stars just happen to appear along almost the same line of sight. The two stars can be at very different distances.

Visual binary. These stars are in orbit about each other and we can see both stars directly.

Composite spectrum binary. When we take a spectrum of the star, we see the lines of two different spectral type stars. From this we infer the presence of two stars.

Eclipsing binary. As we observe the light from such a system, it periodically becomes brighter and fainter. We interpret the dimming as occurring when the companion passes behind or in front of the main star. For us to see these eclipses, we must be aligned in the plane of the orbit (Fig. 5.1). A famous example of such a star is *Algol* (also known as β Persei, indicating that it is the second brightest star in the constellation Perseus).

Astrometric binary. Astrometry is the branch of astronomy in which the positions of objects are measured very accurately. In an astrometric binary, we can only see the brighter star. However, when we follow its path on the sky, we see that, instead of following a straight line, it 'wobbles' back and forth across the straight line path. This means that the star is moving in an orbit, so we can infer the presence of a companion (Fig. 5.2).

Spectroscopic binary. When we study the spectrum of a star we may see that the wavelengths of spectral lines oscillate periodically about the average wavelength. We interpret these variations as being caused by a Doppler shift (discussed in the next section). When the star is coming towards us in its orbit, we see the lines at shorter wavelengths, and when the star is moving away from us in its orbit, we see the lines at longer wavelengths.

It is possible for a given binary system to fit into more than one of these categories, depending on what we can observe.

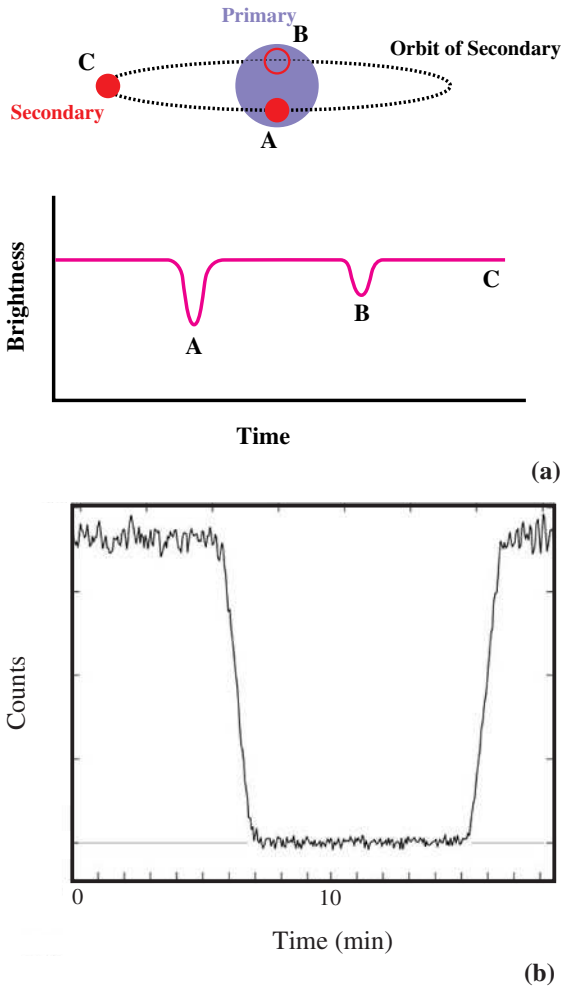

Fig 5.1. Eclipsing binary. (a) The binary system is shown above; the light curve below. The fainter secondary passes alternately in front of and behind the primary. Most of the time, as at position C, we see light from both stars. When the secondary eclipses the primary, part of the primary light is blocked and there is a dip in the intensity, as at point A. When the secondary passes behind the primary, as at B, its light is lost. Since the secondary is not as bright as the primary, the loss of brightness is not as great as at A. (b) Part of the light curve for the eclipsing binary, NN Ser, around the eclipse part of the orbit. [(b) ESO]

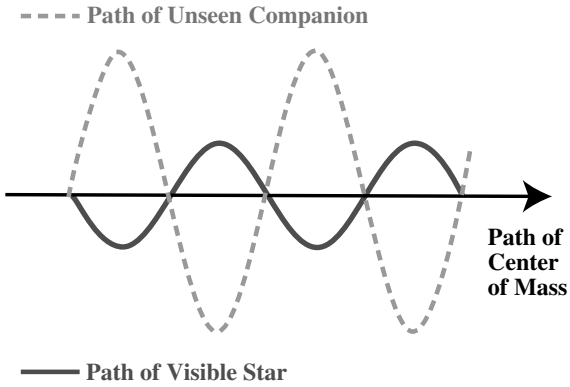

Fig 5.2. Astrometric binary. Two stars orbit about a common center of mass, which in turn moves across the sky. The fainter star is too faint to see, so we only see the brighter star, moving back and forth across the path of the center of mass.

5.2 Doppler shift

A *Doppler shift* is a change in the wavelength (and frequency) of a wave, resulting from the motion of the source and/or the observer. It is most easily visualized for a sound wave or a water wave, where the waves are moving through a particular elastic medium (Fig 5.3).

5.2.1 Moving sources and observers

We first look at the case of the moving observer. If the observer is moving toward the source, then the waves will be encountered more frequently than if the observer were standing still. This means that the observed frequency of the wave increases. If the frequency increases, then the wavelength decreases. If the observer is moving away from the source then the situation is reversed. Waves will be encountered less frequently; the frequency decreases; the wavelength therefore increases. It should be noted that if the observer moves perpendicular to the line joining the source and observer, no shift will be observed.

We now look at the case of the moving source. Each wavefront is now emitted in a different place. If the source is moving toward the observer, the waves will be emitted closer together than if the source were standing still. This means the wavelength decreases. The decreased wavelength results in an increased frequency. If the source is moving away from the observer, the waves will be emitted farther apart than if the source were standing still. The wavelength increases and the frequency decreases. Again, if the source is moving perpendicular to the line joining the source and observer, no shift results.

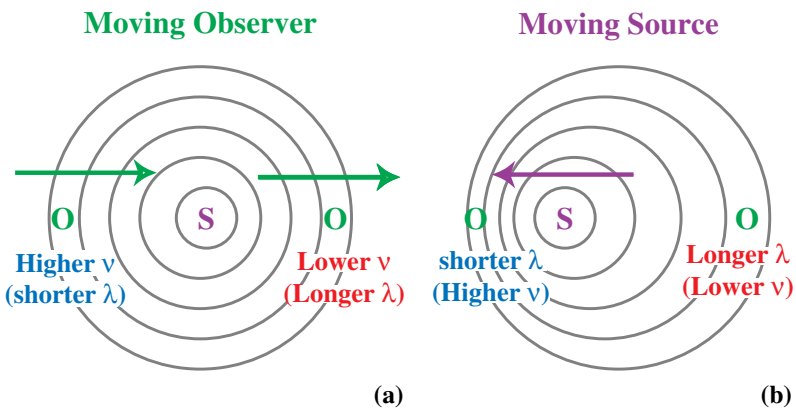

Fig 5.3. Doppler shift for waves in an elastic medium, such as sound waves. (a) Moving observer. On the left the observer is moving toward the source, encountering wave crests more frequently than for a stationary observer. The frequency appears to increase (and the wavelength to decrease). On the right, the observer is moving away from the source, encountering waves at a lower frequency, corresponding to a lower frequency, corresponding to a longer wavelength. (b) Moving source. The motion of the source distorts the wave pattern, so the circles are no longer concentric. The observer on the left has the source approaching, producing a shorter wavelength (and a higher frequency). The observer on the right has the source receding, producing a longer wavelength (and a lower frequency).

It is possible for both the source and the observer to be moving. If their combined motion brings them closer together, the wavelength will decrease and the frequency will increase. If their combined motion makes them move farther apart, the wavelength will increase, and the frequency will decrease. If there is no instantaneous change in their separation, there is no shift in wavelength or frequency.

The shift only depends on the component of the relative velocity along the line joining the source and observer, since this is the only component that can change the distance, r, between them. We call this component the *radial velocity* (Fig. 5.4). We refer to the line joining the source and observer as the *line of sight*. From our definition of radial velocity, v_r, we can see that it is given by

$$v_r = dr/dt \tag{5.1}$$

Note that if the source and observer are moving apart, r is increasing, and $v_r > 0$. If the source and observer are moving together, r is decreasing, and $v_r < 0$.

Suppose the source is moving with a speed v_s in a direction that makes an angle θ with the line of sight, and the observer is moving with a speed v_o in a direction making an angle ϕ with the line of sight. Taking the components of the two velocities along the line of sight as $v_s \cos\theta$ and $v_o \cos\phi$, and subtracting the get the relative radial velocity, gives

$$v_r = v_s \cos\theta - v_o \cos\phi \tag{5.2}$$

In astronomy we are interested in the Doppler shift for electromagnetic waves. The underlying physics is a little different, because there is no mechanical medium for these waves to move through. They can travel even in a vacuum. (We will discuss this point further in Chapter 7.) For sound waves, the actual amount of Doppler shift depends on whether the source or observer (or

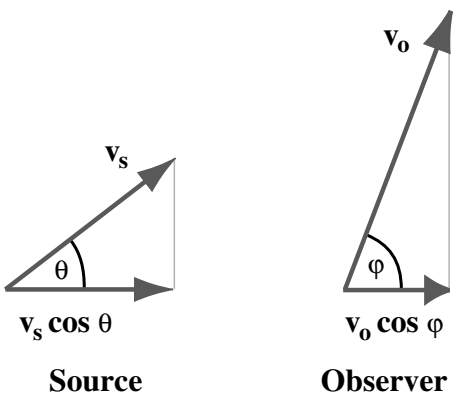

Fig 5.4. Radial velocity. The horizontal line is the line of sight between the source and observer. The radial velocity is the difference between the line of sight components of the observer and source velocities.

both) is moving. For electromagnetic waves, only the relative motion counts.

As long as the relative speed of the source and the observer is much less than the speed of light, the results for electromagnetic radiation are relatively simple. If λ is the wavelength at which a signal is received, and λ_0 is the wavelength at which it was emitted, called the *rest wavelength*, the wavelength shift $\Delta\lambda$ is defined by

$$\Delta\lambda = \lambda - \lambda_0 \tag{5.3}$$

The simple result is that the wavelength shift, expressed as a fraction of the original wavelength, is equal to the radial velocity, expressed as a fraction of the speed of light. That is

$$\Delta\lambda/\lambda_0 = v_r/c \qquad (v_r \ll c) \tag{5.4}$$

If $v_r > 0$, then $\Delta\lambda > 0$. For a spectral line in the middle of the visible part of the spectrum, a shift to longer wavelength is a shift to the red, so this is called a *redshift*. The name applies even if we are in other parts of the spectrum. A positive radial velocity always produces a redshift. If $v_r < 0$, then $\Delta\lambda < 0$, and we have a *blueshift*.

We now look at what happens to the frequency. We remember that $\lambda = c/\nu$, so

$$d\lambda/d\nu = -c/\nu^2$$
$$= -\lambda/\nu$$

This means that $\Delta\nu = (-\nu/\lambda)\,\Delta\lambda$. Substituting into equation (5.4) gives

$$\Delta\nu/\nu_0 = -v_r/c \quad (v_r \ll c) \tag{5.5}$$

The shift in frequency, expressed as a fraction of the rest frequency, is the negative of the radial velocity, expressed as a fraction of the speed of light. (So $\Delta\nu/\nu_0 = -\Delta\lambda/\lambda_0$.)

Example 5.1 Doppler shift

The Hα line has a rest wavelength, $\lambda_0 = 656.28$ nm. What is the observed wavelength for a radial velocity (a) $v_r = 10$ km/s, and (b) $v_r = -10$ km/s?

SOLUTION

(a) We use equation (5.4) to find the wavelength shift:

$$\Delta\lambda = \lambda_0(v_r/c)$$
$$= \frac{(656.28\text{ nm})(10.0\text{ km/s})}{(3.0 \times 10^5\text{ km/s})} = 0.022\text{ nm}$$

We now add this to the original wavelength to find the observed wavelength:

$$\lambda = \lambda_0 + \Delta\lambda$$
$$= 656.28\text{ nm} + 0.022\text{ nm}$$
$$= 656.30\text{ nm}$$

(b) If we take the negative of v_r, we just get the negative of $\Delta\lambda$, so

$$\Delta\lambda = -0.022\text{ nm}$$

This gives a wavelength of

$$\lambda = 656.26\text{ nm}$$

If we observe two spectral lines, their wavelength shifts will be different, since each is shifted by an amount proportional to its own rest wavelength. Thus, the spacing between spectral lines will be shifted.

5.2.2 Circular orbits

We now look at Doppler shifts produced by a star in a circular orbit. The orbital speed is v, and the radius is r. The angular speed of the star in its orbit (in radians per second) is given by $\omega = v/r$. The situation is shown in Fig. 5.5. Suppose the star is moving directly away from the observer at time $t = 0$. At that instant the radial velocity $v_r = v$. As the stars moves, the component of its

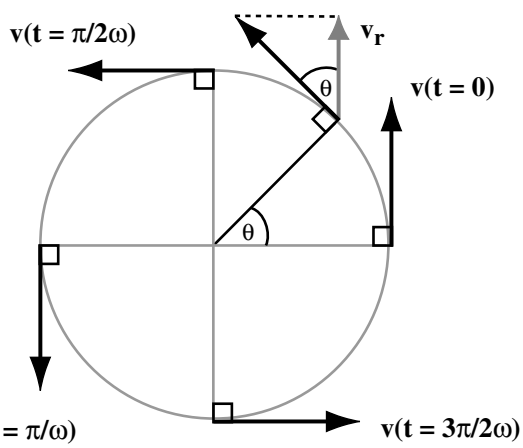

Fig 5.5. Doppler shift for a circular orbit. The speed v of the source remains constant, but the direction changes, so the radial velocity v_r changes. The angle θ keeps track of how far around the circle the source has gone. The source velocity is shown for five different values of θ.

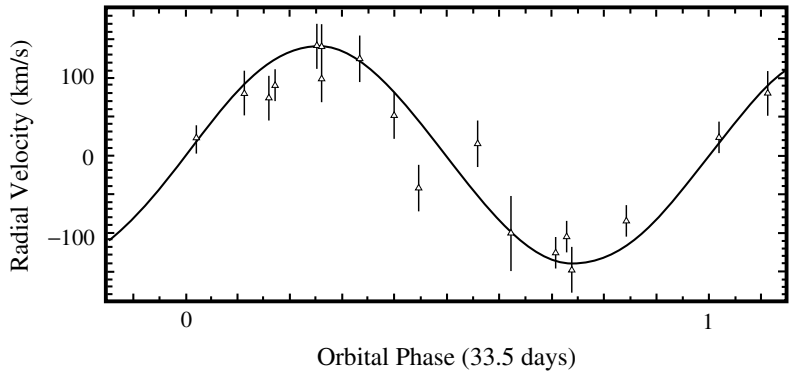

Fig 5.6. Radial velocity curve for the spectroscopic binary GRS1915+105, which has an orbital period of 33.5 days. The horizontal axis is the fraction through that orbit. Data points and error bars are indicated. The smooth curve is the best fit to the data. [ESO]

velocity along the line of sight is $v \cos \theta$. However, $\theta = \omega t$, so

$$v_r = v \cos(\omega t) \tag{5.6}$$

The radial velocity changes sign every half-cycle, and repeats periodically. This is shown in Fig. 5.6. The period of the motion is the circumference, $2\pi r$, divided by the speed v, so $P = 2\pi/\omega$. If we substitute equation (5.6) into equation (5.4), we find that the spectral lines shift back and forth, with a shift given by

$$\Delta\lambda/\lambda_0 = (v/c) \cos(\omega t)$$

So far we have been considering the situation in which the observer is in the plane of the orbit. If the observer is not in the plane of the orbit, the Doppler shift will be reduced (Fig. 5.7). If i is the angle between the plane of the orbit and the plane of the sky, then the projection of any velocity in the plane of the orbit into the line of sight is $v \sin i$. The angle i is known as the *inclination* of the orbit. This gives us a radial velocity for an orbiting star

$$v_r = v \sin i \cos(\omega t) \tag{5.7}$$

5.3 Binary stars and circular orbits

In this section we will see how Newton's laws of motion and gravitation can be applied to binary stars in circular orbits. Circular orbits are not the most general case of orbital motion, but the analysis is most straightforward, and most of the basic points are clearly illustrated. In the next section we will go to the general case of elliptical orbits.

We consider two stars, of masses m_1 and m_2, orbiting their common center of mass at distances r_1 and r_2, respectively (Fig. 5.8). From the

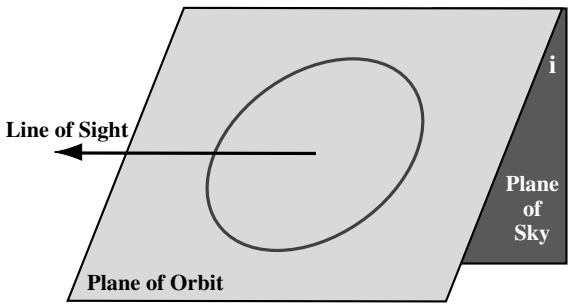

Fig 5.7. Inclination of an orbit. The orbit is an ellipse, which lies in a plane. The plane makes some angle i with a plane of the sky. The plane of the sky is defined to be perpendicular to the line of sight.

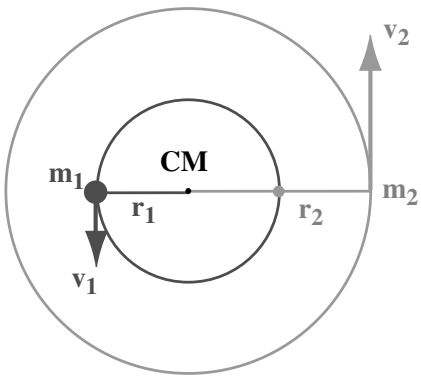

Fig 5.8. Binary system with circular orbits. Both stars orbit the center of mass (CM). The more massive star is closer to the center of mass. The center of mass must always be between the two stars, so the stars lie on opposite sides of it.

definition of center of mass, these quantities are related by

$$m_1 r_1 = m_2 r_2 \tag{5.8}$$

The center of mass moves through space subject only to the external forces on the binary star system. The forces between the two stars do not affect the motion of the center of mass. We will therefore ignore the actual motion of the center of mass, and view the situation as it would be viewed by an observer sitting at the center of mass.

Since the center of mass must always be along the line joining the two stars, the stars must always be on opposite sides of the center of mass. This means that the stars orbit with the same orbital period P. In general, the period of the orbit is related to the radius, r, and the speed, v, by

$$P = 2\pi r/v \tag{5.9}$$

Solving for v gives

$$v = 2\pi r/P \tag{5.10}$$

Since the periods of the two stars must be the same, equation (5.9) tells us that

$$r_1/v_1 = r_2/v_2 \tag{5.11}$$

Combining these with equation (5.8) gives

$$v_1/v_2 = r_1/r_2 = m_2/m_1 \tag{5.12}$$

(Note: We could have also obtained $m_1 v_1 = m_2 v_2$ directly from conservation of momentum. This is not surprising since the properties of the center of mass come from conservation of momentum.)

We now look at the gravitational forces. The distance between the two stars is $r_1 + r_2$, so the force on either star is given by Newton's law of gravitation as

$$F = G\frac{m_1 m_2}{(r_1 + r_2)^2} \tag{5.13}$$

This force must provide the acceleration associated with the change of the direction of motion in circular motion, v^2/r. For definiteness, we look at the force on star 1, so

$$F = m_1 v_1^2/r_1 \tag{5.14}$$

Combining equations (5.13) and (5.14) gives

$$\frac{m_1 v_1^2}{r_1} = G\frac{m_1 m_2}{(r_1 + r_2)^2} \tag{5.15}$$

Note that we can divide both sides by m_1. If we also use equation (5.10) to relate v_1 to P, we find that

$$\frac{4\pi^2 r_1}{P^2} = \frac{G\, m_2}{(r_1 + r_2)^2} \tag{5.16}$$

This can be simplified if we introduce the total distance R between the two stars:

$$R = r_1 + r_2$$

$$= r_1(1 + r_2/r_1) \tag{5.17}$$

Using equation (5.12), this becomes

$$R = r_1(1 + m_1/m_2) \tag{5.18}$$

$$= (r_1/m_2)(m_1 + m_2) \tag{5.19}$$

Substituting this into equation (5.16) gives

$$4\pi R^3/G = (m_1 + m_2)P^2 \tag{5.20}$$

Let's look at how equation (5.20) can be used to give us stellar masses. For any binary system, we can determine the period directly if we watch the system for long enough. If the star is a spectroscopic binary, we can see how long it takes for the Doppler shifts to go through a full cycle. If it is an astrometric binary we can see how long it takes for the 'wobble' to go through a full cycle. If it is an eclipsing binary, we can see how long it takes the light curve to go through a full cycle. If we can see both stars, we can determine R. Once we know R and P, we can use equation (5.20) to determine the sum of the masses, $(m_1 + m_2)$. We can also obtain the ratio of the masses m_1/m_2, either from r_1/r_2, if both stars can be seen, or v_1/v_2, if both Doppler shifts are observed. Once we know the sum of the masses and the ratio of the masses, the individual masses can be determined. The situation we have outlined here is the ideal one, however. Usually, we don't have all of these pieces of information (as we will see below).

Example 5.2 Mass of the Sun

We can consider the Sun and Earth as a binary system, so we should be able to apply equation (5.20) to

find the mass of the Sun. It turns out that this is the most accurate measure we have of the Sun's mass.

SOLUTION

Since the mass of the Sun is so much greater than that of the Earth, we can approximate the sum of the masses as being the mass of the Sun, $M_\odot$. Equation (5.20) then becomes

$$M_\odot = \frac{4\pi^2 R^3}{GP^2}$$

$$= \frac{4\pi^2(1.5 \times 10^{13}\ \text{cm})^3}{(6.67 \times 10^{-8}\ \text{dyn cm}^2/\text{g}^2)(3.16 \times 10^7\ \text{s})^2}$$

$$= 2 \times 10^{33}\text{g}$$

We call this quantity a *solar mass*. It becomes a convenient quantity for expressing the masses of other stars.

From the Earth and Sun we know that for a pair of objects orbiting with a period of 1 yr, at a distance of 1 AU (defined in Section 2.6), the sum of the masses must be one solar mass. This suggests a convenient system of units for equation (5.20). If we express masses in solar masses, distances in astronomical units, and periods in years, the constants must equal one to give the above result. We can therefore rewrite equation (5.20) as

$$\left[\frac{R}{1\ \text{AU}}\right]^3 = \left[\frac{m_1 + m_2}{1\ M_\odot}\right]\left[\frac{P}{1\ \text{yr}}\right]^2 \tag{5.21}$$

For the Solar System, we write the sum of the masses as one in these units, so the equation simply says that the cube of the radius (in AU) is equal to the square of the period (in yr). This is also known as *Kepler's third law of planetary motion*. The law was originally found by Kepler observationally, and Newton used it to show that gravity must be an inverse square law force. (See Problem 5.10.) We will discuss planetary motions in more detail in Chapter 22.

For a visual binary, we don't directly measure R, the linear separation. We actually measure the angular separation on the sky, θ. If d is the distance to the binary, then R is equal to $\theta(\text{rad})d$, where $\theta(\text{rad})$ is the value of θ measured in radians. When we use this relation, R and d will come out in the same units. Therefore

$$R(\text{AU}) = d(\text{AU})\theta(\text{rad})$$

The values of θ are so small that radians are an inconvenient quantity. We can convert to arc seconds (equation (2.16)) to give

$$R(\text{AU}) = d(\text{AU})\theta('')(2.06 \times 10^5)$$

The factor of 2.06×10^5 was to convert radians to arc seconds, but it is also the factor to convert astronomical units to parsecs, so we have

$$R(\text{AU}) = d(\text{pc})\theta('')$$

If we use equation (2.17) to relate the distance in parsecs to the parallax in arc seconds, this becomes

$$R(\text{AU}) = \theta('')/p('') \tag{5.22}$$

This can then be substituted directly into equation (5.21).

We will now look at the behavior of the Doppler shifts. Applying equation (5.10) to both speeds, v_1 and v_2, and remembering that the period of the orbit is the same for both stars, we have

$$r_1 + r_2 = (P/2\pi)(v_1 + v_2)$$

Using this to eliminate R in equation (5.20) gives

$$(P/2\pi G)(v_1 + v_2)^3 = m_1 + m_2 \tag{5.23}$$

If the orbit is inclined at an angle i, then the Doppler shifts only measure the components $v_r = v\sin(i)$. In terms of the radial velocities v_{1r} and v_{2r}, equation (5.23) becomes

$$(P/2\pi G)(v_{1r} + v_{2r})^3/\sin^3 i = m_1 + m_2 \tag{5.24}$$

If a binary happens to be an eclipsing binary, then we know that we are close to the plane of the orbit, and i is close to 90°. Otherwise we don't know i. If a circular orbit is projected at some angle on the plane of the sky it will appear elliptical. We will see in the next section that there are ways to determine i if we can trace that projected orbit on the sky. If i is unknown all we can do is solve equation (5.24) with $i = 90°$. This will give us a value of $m_1 + m_2$ that is a *lower limit* to the true value. The true value would be this lower limit divided by $\sin^3 i$, and since $\sin^3 i$ is less than or equal to unity, the value assuming $i = 90°$ is less than the true value. Finding lower limits is not as useful as finding actual values. However, if we study enough binary systems, we will encounter a full range of inclination angles.

These statistical studies can be used to relate mass to spectral type.

Example 5.3 Binary star Doppler shifts
A binary system is observed to have a period of 10 yr. The radial velocities of the two stars are determined to be $v_{1r} = 10$ km/s and $v_{2r} = 20$ km/s, respectively. Find the masses of the two stars (a) if the inclination of the orbit is 90° , and (b) if it is 45°.

SOLUTION
From equation (5.24), we have

$$\frac{m_1 + m_2}{M_\odot} = \frac{(10\text{ yr})(3.16 \times 10^7\text{ s/yr})[(10 + 20)(1 \times 10^5\text{ cm/s})]^3}{2\pi(6.67 \times 10^{-8}\text{ dyn cm}^2/\text{g}^2)(2 \times 10^{33}\text{ g})(\sin^3 i)}$$

$$= 10.2\, M_\odot/\sin^3 i$$

This means that

$$m_1 + m_2 = 10.2\, M_\odot/\sin^3 i$$

If $i = 90°$, $\sin^3 i = 1$, so

$$m_1 + m_2 = 10.2\, M_\odot$$

We find the ratio of the masses from the ratio of the radial velocities

$$m_1/m_2 = v_2/v_1$$
$$= 2.0$$

This means that $m_1 = 2m_2$, so

$$2m_2 + m_2 = 10.2\, M_\odot$$

giving $m_1 = 6.8\, M_\odot$ and $m_2 = 3.4\, M_\odot$. If $i = 45°$, $1/\sin^3 i = 2.8$. The ratio of the masses does not change, since the sin i drops out of the ratio of the radial velocities. This means that we can just multiply each mass by 2.8 to give 19.2 and 9.5 $M_\odot$, respectively.

It is often the case that only one Doppler shift can be observed. Let's assume that we measure v_1 but not v_2. We must therefore eliminate v_2 from our equations. We can write v_2 as

$$v_2 = v_1(m_1/m_2)$$

The sum of the velocities then becomes

$$v_1 + v_2 = v_1(1 + v_2/v_1)$$
$$= v_1(1 + m_1/m_2)$$
$$= (v_1/m_2)(m_1 + m_2)$$

If we now substitute into equation (5.24), we find

$$\left(\frac{P}{2\pi G}\right)(v_{ir})^3 = \frac{m_2^3 \sin^3 i}{(m_1 + m_2)^2} \tag{5.25}$$

The quantity on the right-hand side of equation (5.25) is called the *mass function*. If we can measure only one Doppler shift, we cannot determine either of the masses. We can only measure the value of the mass function. We can, however, obtain information on the masses of various spectral types through extensive statistical studies.

We can also look at energy of a binary system with circular orbits. It is the sum of the kinetic energies of the two stars plus the gravitational potential energy, which we take to be zero when the two objects are infinitely far apart. The energy is

$$E = (1/2)m_1 v_1^2 + (1/2)m_2 v_2^2 - Gm_1 m_2/R \tag{5.26}$$

From equation (5.15), we have

$$m_1 v_1^2 = Gr_1 \frac{m_1 m_2}{(R)^2}$$

and, using equation (5.12), we obtain

$$m_2 v_2^2 = Gr_2 \frac{m_1 m_2}{(R)^2}$$

Substituting these into equation (5.26) and simplifying gives

$$E = -(1/2)\, Gm_1 m_2/R \tag{5.27}$$

Compare this with equation (3.4), which has the energy for circular orbits with electrical (rather than gravitational) forces.

The negative energy means that the system is bound. We would have to add at least $(1/2)Gm_1m_2/R$ to break up the binary system. (We can think of this as being analogous to the binding energy of a hydrogen atom.) For any pair of masses, as you make the orbits of the binary system smaller, the energy becomes more negative.

Example 5.4 Binding energy of a binary system.
What is the binding energy of a binary system with two 1 $M_\odot$ stars orbiting with each 100 AU from the center of mass?

SOLUTION

Using equation (5.27)

$$E = -\left(\frac{1}{2}\right)\frac{Gm_1m_2}{R}$$

$$= -\left(\frac{1}{2}\right)\frac{(6.67 \times 10^{-8}\text{dyn cm}^2/\text{g}^2)(2 \times 10^{33}\,\text{g})^2}{(200)(1.5 \times 10^{13}\text{cm})}$$

$$= -4.5 \times 10^{43}\ \text{erg}$$

5.4 Elliptical orbits

5.4.1 Geometry of ellipses

In general, orbiting bodies follow elliptical paths. A circle is just a special case of an ellipse. In this section, we generalize the results from the previous section from circular orbits to elliptical orbits. The basic underlying physical ideas are the same.

We first review the geometry of an ellipse, as shown in Fig. 5.9. We describe the ellipse by its semi-major axis a and semi-minor axis b. Each point on the ellipse satisfies the condition that the sum of the distances from any point to two fixed points, called *foci* (singular *focus*), is constant. If r and r' are these two distances then

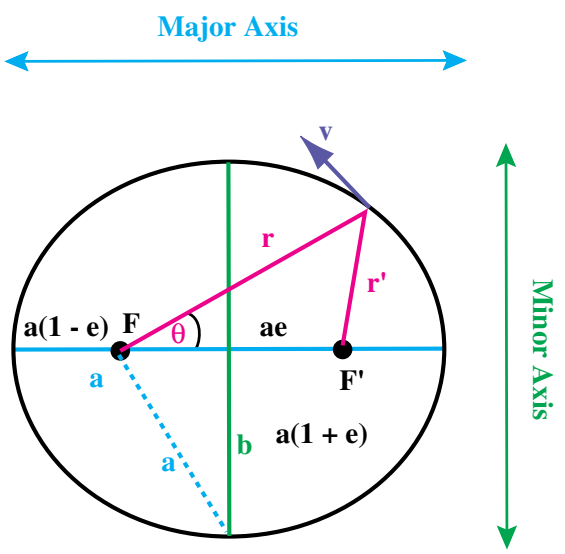

Fig 5.9. Geometry of an ellipse. The length of the semi-major axis is a; the length of the semi-minor axis is b. The two foci are at F and F'. The eccentricity is e, and the distances from the two foci to points on the ellipse are r and r'.

$r + r'$ is a constant. We can see that for a point on the semi-major axis (and the ellipse), this sum is $2a$, so it must be $2a$ everywhere. That is

$$r + r' = 2a \tag{5.28}$$

The *eccentricity* of an ellipse is the distance between the foci, divided by $2a$. A circle is an ellipse of eccentricity zero (both foci are at the same point, the center of the circle). The eccentricity of any ellipse must be less than unity. From the point where the curve crosses the minor axis, $r = r' = a$, so

$$b^2 = a^2 - (ae)^2$$

$$= a^2(1 - e^2) \tag{5.29}$$

In a binary system, the center of mass of the two stars will be at one focus on the ellipse. The farthest point from that focus is called the *apastron*. From the figure, we see that the distance from the focus to this point is

$$r(\text{apastron}) = a(1 + e) \tag{5.30a}$$

The closest point to the focus is called the *periastron*. Its distance from the focus is

$$r(\text{periastron}) = a(1 - e) \tag{5.30b}$$

The average of these two values is a, the semi-major axis. This is the quantity that replaces the radius of a circular orbit in our study of binary stars.

It is useful to have an expression for the ellipse, relating the variables r and θ. From the law of cosines, we see that

$$r'^2 = r^2 + (2ae)^2 + 2r(2ae)\cos\theta \tag{5.31}$$

Using equation (5.28) gives

$$r = \frac{a(1 - e^2)}{1 + e\cos\theta} \tag{5.32}$$

5.4.2 Angular momentum in elliptical orbits

The gravitational force between two objects always acts along the line joining the two objects. The center of mass also lies along this line. This means that the force on either object points directly from that object towards the center of mass. Therefore, these forces can exert no torques about the center of mass. If there are no torques about the center of

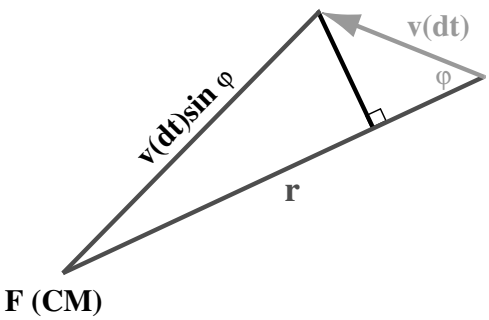

Fig 5.10. Angular momentum in an elliptical orbit. In this case the object is a distance r from the focus F (which is also the center of mass). Its velocity makes an angle ϕ with the line from F to the object. (If this were a circle, ϕ would be 90°.) The time interval over which we mark the motion is dt.

mass, then the angular momentum about the center of mass is conserved.

To consider the angular momentum of an object in an elliptical orbit, we look at Fig. 5.10. For an object of mass m, the angular momentum about the center of mass (which corresponds to one of the foci) is r times the component of its linear momentum perpendicular to the direction of r. That is

$$L = mvr \sin \phi \tag{5.33}$$

where ϕ is between the line from the center of mass to the object (of length r) and its direction of motion (the direction of the velocity vector, $\boldsymbol{v}$).

We look at the area swept out by the line from the center of mass to r, in the time interval dt. The area is the thin triangle shown in the figure. The long side of the triangle is r and the short side is $v\,\mathrm{d}t$. The small right triangle shows that the height of the larger triangle is $v\,(\mathrm{d}t)\sin\phi$. The area is then half the base × height, so

$$\mathrm{d}A = (1/2)\, r(v\,\mathrm{d}t)\sin\phi$$

The rate at which the area is swept out is

$$\mathrm{d}A/\mathrm{d}t = (1/2) r\, v \sin\phi$$

However, we can use equation (5.33) to eliminate $r\,v\sin\varphi$, giving

$$\mathrm{d}A/\mathrm{d}t = (1/2)\,(L/m)$$

Since L is constant, the rate at which area is swept out is constant. Equal areas are swept out in equal times. (When applied to planetary motions, this is known as *Kepler's second law*, as we will discuss in Chapter 22.)

The major consequence of angular momentum conservation is therefore that the objects move slower when they are farther apart and faster when they are closer together. (As we will see, the Earth's orbit is only slightly eccentric but it moves faster when it is closer to the Sun, in January, and slower when it is farther away. This results in winter and spring in the northern hemisphere being shorter than summer and winter. Check a calendar to see this.)

5.4.3 Energy in elliptical orbits

We next look at the total energy of the binary system. Adding the kinetic energies of the two stars and their gravitational potential energy (defined as zero when the stars are infinitely far apart) gives

$$E = (1/2)m_1 v_1^2 + (1/2)\, m_2 v_2^2 - Gm_1 m_2/R \tag{5.34}$$

In this expression v_1 and v_2 are the speeds relative to the center of mass. Remember, from the definition of the center of mass, we found that $m_1 v_1 = m_2 v_2$. We introduce the relative speed of the two stars

$$v = v_1 + v_2 \tag{5.35a}$$

The two speeds are added because the two stars are always moving in opposite directions, so their relative speed will be the sum of the magnitudes of their individual speeds. In terms of v, the two speeds are

$$v_1 = \frac{m_2 v}{m_1 + m_2} \tag{5.35b}$$

$$v_2 = \frac{m_1 v}{m_1 + m_2} \tag{5.35c}$$

Substituting into equation (5.32) gives

$$E = m_1 m_2 \left[\frac{v^2}{2(m_1 + m_2)} - \frac{G}{R} \right] \tag{5.36}$$

Since energy is conserved, we can evaluate it at any point we want. For simplicity, we choose apastron (speed v_a) and periastron (speed v_p). We can relate the speeds v_a and v_p by conservation of angular momentum. Angular momentum conservation is easy to apply at the apastron and periastron because the velocities are perpendicular to the line from the focus to the star. Since $\phi = 90°$ at these points, the angular momentum (equation (5.33)) is just mvr. Using equations (5.30a) and

(5.30b) we find that

$$a(1 + e)v_a = a(1 - e)v_p \tag{5.37}$$

Solving for the ratio v_p/v_a gives

$$\frac{v_p}{v_a} = \frac{1 + e}{1 - e} \tag{5.38}$$

Now that we have the ratio v_p/v_a, we need another relation between them to be able to solve for v_a and v_p individually. We can use conservation of energy to equate the energies at the apastron and periastron. Using equation (5.36) gives

$$\frac{v_a^2}{2(m_1 + m_2)} - \frac{G}{a(1 + e)} = \frac{v_b^2}{2(m_1 + m_2)} - \frac{G}{a(1 - e)} \tag{5.39}$$

Rearranging gives

$$\left[\frac{G(m_1 + m_2)}{a}\right]\left(\frac{1}{1 - e} - \frac{1}{1 + e}\right) = \frac{1}{2}(v_p^2 - v_a^2)$$

$$= \frac{1}{2}v_a^2\left[\left(\frac{v_p^2}{v_a^2}\right) - 1\right]$$

We can use equation (5.38) to eliminate the ratio v_p/v_a. Solving for v_a gives

$$v_a^2 = \left[\frac{G(m_1 + m_2)}{a}\right]\left(\frac{1 - e}{1 + e}\right) \tag{5.40a}$$

$$v_b^2 = \left[\frac{G(m_1 + m_2)}{a}\right]\left(\frac{1 + e}{1 - e}\right) \tag{5.40b}$$

If we put these into equation (5.36), the total energy simplifies to

$$E = -Gm_1m_2/2a \tag{5.41}$$

We can now use this in the left-hand side of equation (5.36). We can then solve for v at any point r:

$$v^2 = G(m_1 + m_2)(2/r - 1/a) \tag{5.42}$$

5.4.4 Observing elliptical orbits

In studying the Doppler shifts of elliptical orbits as compared with circular orbits, there are three important differences:

(1) The speed along an elliptical orbit is not constant.
(2) In an elliptical orbit the velocity is not perpendicular to the line from the center of mass to the orbiting object.
(3) Even if you are in the plane of the orbit, the radial velocity curve depends on where you are relative to the major axis of the ellipse.

These points are illustrated in Fig. 5.11.

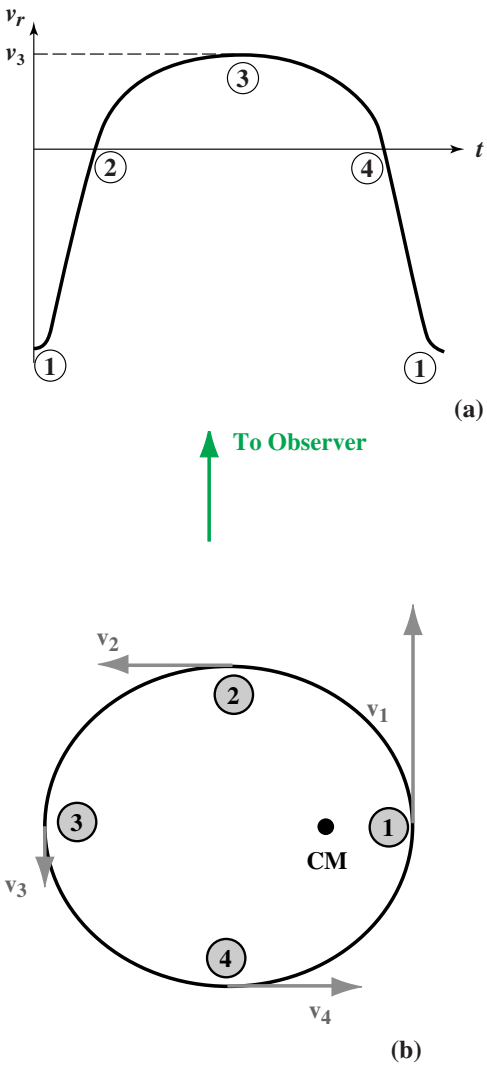

Fig 5.11. (a) Radial velocity vs. time, t, for an elliptical orbit. (b) In contrast to the circular orbit both the magnitude and direction of $\boldsymbol{v}$ change throughout the orbit. (We assume for this figure that the observer is in the plane of the orbit, or that $i = 90°$.) Four points are shown in the orbit and in the radial velocity curve. At points 2 and 4 the motion is perpendicular to the line of sight, so $v_r = 0$. For point 1 the motion is directly toward the observer, producing the maximum negative v_r, and, for point 3, the motion is away from the observer, producing the maximum positive v_r. The motion is also faster at 1 than at 3. In addition, going from 4 to 1 to 2 takes less time than going from 2 to 3 to 4. This accounts for the distorted shape of the radial velocity curve.

As we said above, we must correct any Doppler shift for the inclination of the orbit, i. If we take a tilted ellipse and project it onto the sky, we still have an ellipse. However, that ellipse will have a different eccentricity than the true ellipse. When we look at an elliptical orbit, how can we tell if it is tilted or not? For a tilted orbit the foci will not appear in the right place for the projected ellipse. Therefore, if we see two stars orbiting a point different from the center of mass, we will know that the orbit is inclined. We can determine the inclination from the degree to which the foci appear to be displaced. In this technique, we don't actually know where the center of mass is, so a process must be used in which we try one position for the center of mass and try to match the projected orbit, repeating the process until a good fit is achieved.

5.5 Stellar masses

As a result of studying many binary systems, astronomers have a good idea of the masses of main sequence stars. These results are summarized in Table 5.1. Just as the Sun's temperature places it in the middle of the main sequence, its mass is in the middle of the range of stellar masses. The lowest mass main sequence stars have about 0.07 of a solar mass, and the most massive stars commonly encountered have about 60 solar masses. When we think of how large or small stars might have turned out to be, the observed range of stellar masses is not very large. This range is an important constraint on theories of stellar structure.

Table 5.1. Mass and spectral type (MS).

Spectral type	$M/M_\odot$
O5	40.0
B5	7.1
A5	2.2
F5	1.4
G5	0.9
K5	0.7
M5	0.2

An even more stringent constraint is the relationship between mass and temperature on the main sequence. The cooler stars are less massive and the hotter stars are more massive. We have already said that the existence of the main sequence implies a certain relationship between size and temperature. This means that if a star is on the main sequence, once its mass is specified, its radius and temperature are determined. Another way of looking at this to say that a star's mass determines where on the main sequence it will fall.

Since the mass determines the radius and temperature of a main sequence star, it should not be surprising that it also determines the luminosity. The exact dependence of the luminosity on mass is called the *mass–luminosity relationship*. This relationship is also explainable from theories of stellar structure. This relationship is shown in Fig. 5.12. We can summarize it by saying that the luminosity varies approximately as some power, α, of the mass. If we express luminosities in terms of solar luminosities, and masses in terms of solar masses, this means that

$$L/L_\odot = (M/M_\odot)^\alpha \tag{5.43a}$$

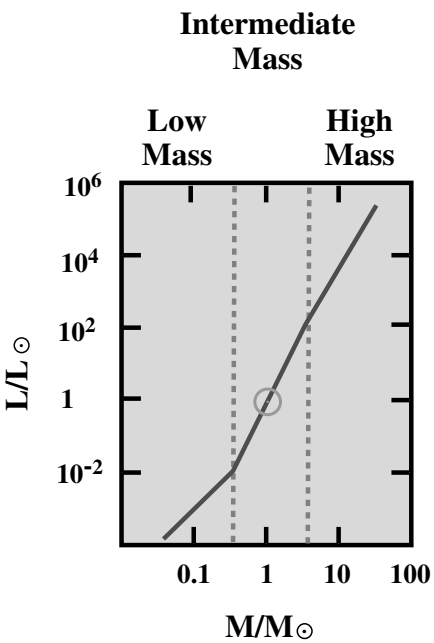

Fig 5.12. Mass–luminosity relationship.

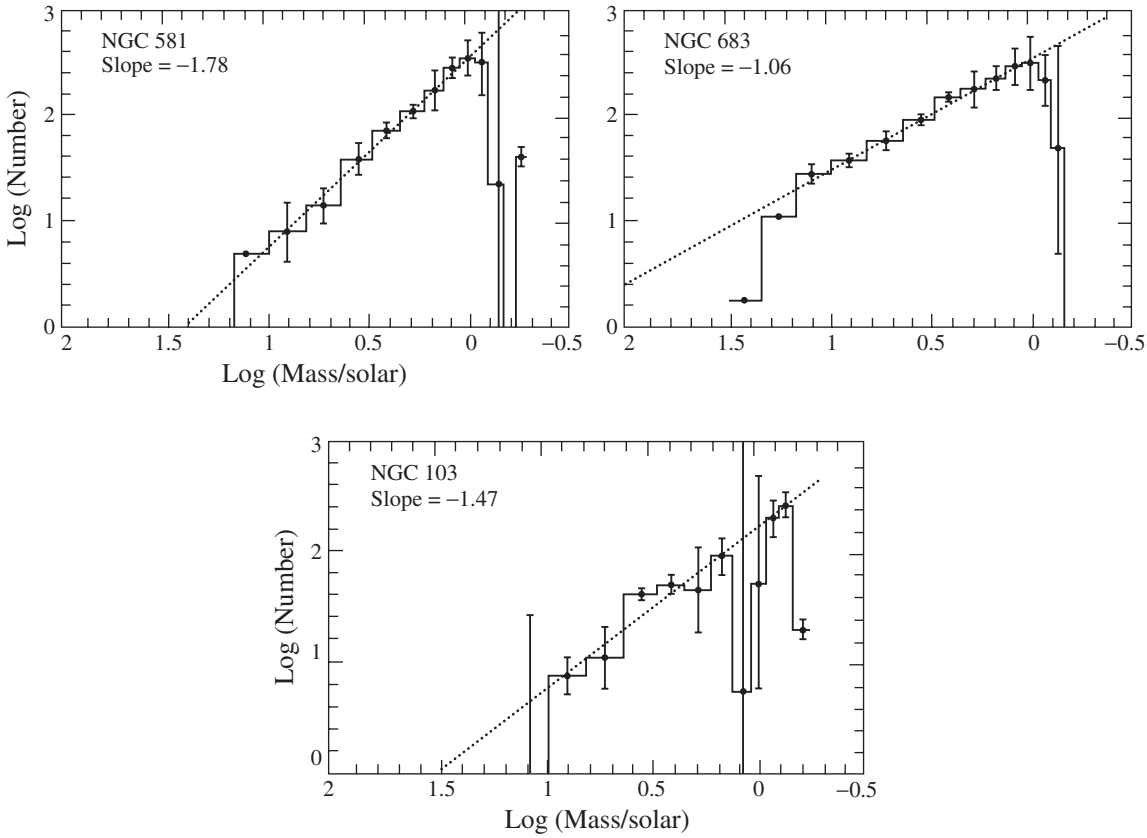

Fig 5.13. Initial mass function for three young clusters. Note that masses are in solar masses, so log(mass) = 0 corresponds to one solar mass. [John Scalo, University of Texas, Austin]

A single value of α does not work for the full range of masses along the main sequence. The approximate values are:

$$\alpha = 1.8 \text{ for } M < 0.3\, M_\odot \qquad \text{(low mass)}$$

$$\alpha = 4.0 \text{ for } 0.3\, M_\odot < M < 3\, M_\odot \qquad \text{(intermediate mass)} \tag{5.43b}$$

$$\alpha = 2.8 \text{ for } 3\, M_\odot < M \qquad \text{(high mass)}$$

In understanding how stars are formed (Chapter 14) we would like to know the distribution of stellar masses. That is, we would like to know the proportions of stars of various masses. Since we now know how to relate mass and spectral type, we can carry out such studies by looking at the relative numbers of different spectral types or luminosities. These studies are difficult, because we can see brighter stars to greater distances than faint stars. This is called a *selection effect*, because it makes it difficult to select an unbiased sample.

It is of interest to study the number of stars in different mass ranges. This is called the mass function. This is a good check on various theories of star formation, which we will discuss further in Chapter 15. When we discuss stellar evolution (Chapters 10 and 11), we will see that low mass stars live longer than high mass stars. So, if we study the mass function in a group of stars (usually clusters, as we will discuss in Chapter 13), it will change with time. As the higher mass stars die, they leave behind a cluster with a depleted number of high mass stars. So, if we want to really see what the mix of masses are when the cluster forms, we need to look at young clusters. This allows us to study what we call the *initial mass function (IMF)*. Data for some sample clusters are shown in Fig. 5.13. Note: the number of low

mass stars is so much greater than the number of high mass stars, that we show the results on a log–log plot. In general, we find that we can fit the data with power laws of the form $N(m) \sim m^{-\gamma}$. The best fit values of γ are shown as slopes in the figure. These figures don't have much data for stars much less massive than the Sun, but other studies indicate that the function doesn't rise as fast as it does for the mass range shown in Fig. 5.13.

We can also look at the number of stars in different luminosity ranges, or the *luminosity function*. We find that there are also many more low luminosity stars than high luminosity stars. However, the luminosity of each high mass star is so much greater than that for each low mass star, that most of the luminosity of our galaxy comes from a relatively small number of high mass stars.

5.6 Stellar sizes

We have alluded so far to stellar sizes, but we have not discussed how they are determined. In this section, we will look at various methods for measuring stellar radii.

The star whose size is easiest to measure is the Sun. This is actually quite useful. We have seen that the Sun is intermediate in its mass and temperature, so its radius is probably a fairly representative stellar radius. The angular radius of the Sun, $\Delta\theta$, is 16 arc min. The Sun is at a distance $d = 1.50 \times 10^8$ km, so its radius, $R_\odot$, is given by

$$R_\odot = d \tan \Delta\theta$$
$$= 6.96 \times 10^5 \text{ km}$$

The Sun is the only star whose disk subtends an angle larger than the seeing limitations of ground-based telescopes. We therefore need other techniques for determining radii. If we know the luminosity (from its absolute magnitude) and the surface temperature (from the spectral type) of a star, we can calculate its radius using equation (2.7). Solving for the radius gives

$$R = (L/4\pi\,\sigma T^4)^{1/2} \tag{5.44}$$

Example 5.5 Luminosity radius

Estimate the radius of an A0 star. (Use Appendix E for the stellar properties.)

SOLUTION

We can express the various quantities in solar units. Taking ratios, we can use equation (5.44) to give

$$R/R_\odot = (L/L_\odot)^{1/2}(T/T_\odot)^{-2}$$
$$= (80)^{1/2}(1.69)^{-2}$$
$$= 3.1$$

Eclipsing binaries (such as in Fig. 5.1) provide us with another means of determining stellar radii. This method involves analysis of the shape of the light curve and a knowledge of the orbital velocities from Doppler shift measurements. (In an eclipsing binary, we don't have to worry about the inclination of the orbit.) Particularly important is the rate at which the light level decreases and increases at the beginning and end of eclipses.

We can also estimate the radii of rotating stars. If there are surface irregularities, such as hot spots or cool spots, the brightness of the star will depend on whether these spots are facing us or are turned away from us. The brightness variations give us the rotation period P. From the broadening of spectral lines, due to the Doppler shift, we can determine the rotation speed v. This speed is equal to the circumference $2\pi R$, divided by the period. Solving for the radius gives

$$R = P\,v/2\pi \tag{5.45}$$

Sometimes the Moon passes in front of a star bright enough and close enough for detailed study. An analysis of these *lunar occultations* tells us about the radius of the star. The larger the star is, the longer it takes the light to go from maximum value to zero as the lunar edge passes in front of the star. Actually, since light is a wave, there are diffraction effects as the starlight passes the lunar limb. The light level oscillates as the star disappears. The nature of these oscillations tells us about the radius of the star.

There is another observational technique, called *speckle interferometry*, that has been quite successful recently. If it were not for the seeing fluctuations in the Earth's atmosphere, we would be able to obtain images of stellar disks down to the diffraction limits of large telescopes. However, the atmosphere is stable for short periods of

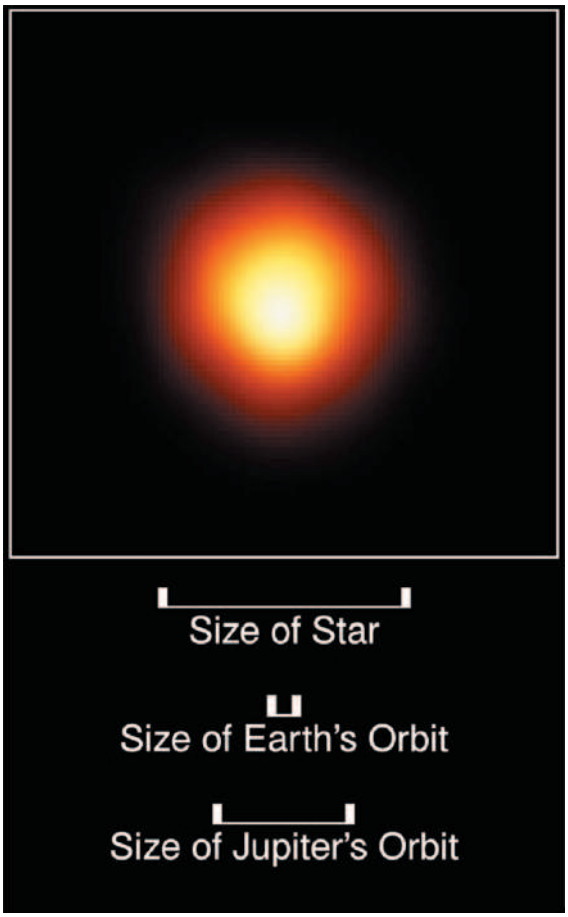

Fig 5.14. Hubble Space Telescope image of the red giant star Betelgeuse. You can see that it is barely resolved, but by understanding how the telescope response smears the image, we can achieve a very accurate estimate for the size of the star. [STScI/NASA]

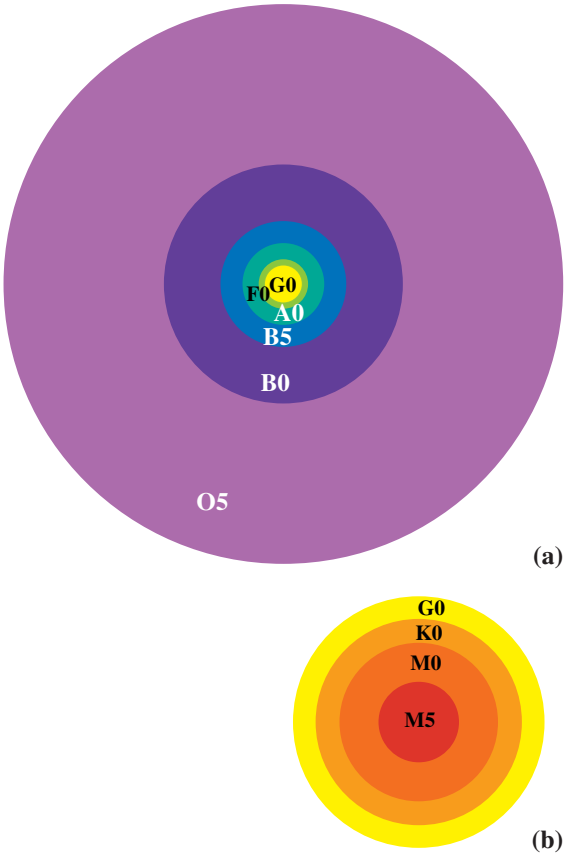

Fig 5.15. Stellar sizes for different spectral types. (a) The largest, down to G0. (b) The smaller ones, from G0 down, but blown up. The G0 circle is repeated to give the relative scale for the two parts of the figure.

time, of the order of 0.01 s. The blurring of images comes from trying to observe for longer than this. If an image were bright enough to see in this short time, we could take a picture with diffraction-limited resolution. Unfortunately, 0.01 s is not long enough to collect enough photons from a star. However, we can collect a series of 0.01 s images, observing interference between light coming through slightly different paths in the atmosphere. The final images must be reconstructed mathematically.

Even more recently, work on stellar sizes has been done on the Hubble Space Telescope, as shown in Fig. 5.14. The results of these various techniques are shown in Fig. 5.15. We can see that, on the main sequence, stars become larger with increasing surface temperature. This is why the luminosity of stars increases with increasing surface temperature at a rate greater than T^4.

Chapter summary

We saw in this chapter how the gravitational interactions in binary systems can be studied to determine the masses of stars.

We saw how the Doppler shift can be used to determine the radial velocities of objects. In binary systems, we can observe the wavelengths

of spectral lines varying periodically as the stars orbit the center of mass. We saw what information could be obtained even if we don't know the orbital inclination.

We saw how the masses of orbiting objects are related to the orbital radii and speeds, and the period of the system for circular orbits. We extended these ideas to elliptical orbits. For elliptical orbits, we saw how conservation of angular momentum means a change in the speed with distance from the center of mass, and how this affects the Doppler shift we see in different parts of the cycle. We also found the total energy for elliptical orbits, and saw how the kinetic energy varied as the speed varies.

We saw that the range of masses for stars along the main sequence is much less than the range of luminosities. There is also a close relationship between mass and luminosity for main sequence stars.

We saw how eclipsing binaries can be used to tell us something about stellar sizes. We also looked at other techniques for determining stellar sizes, including knowing the luminosity and temperature, using lunar occultations and speckle interferometry.

Questions

5.1. Under what conditions can we determine the masses of both stars in a binary system? Think of as many combinations of situations as you can.

5.2. For a binary that is *only* detected as an astrometric binary, what are the conditions under which we can determine the masses of both stars?

5.3. If you observe two stars close together on the sky, how would you decide if they were an optical double or a true binary?

5.4. If we see an eclipsing binary, how do we know the inclination of the orbit?

5.5. Describe situations in which a source and/or observer are moving and no Doppler shift for electromagnetic waves is observed.

5.6. When you hear the Doppler shift in a train whistle, the effect is most obvious as the train goes by you. Why is that?

5.7. For the following situations, indicate whether the radial velocity is positive, negative or zero:
(a) Observer moving towards (stationary) source.
(b) Source moving towards (stationary) observer.
(c) Source moving away from (stationary) observer.
(d) Source and observer moving towards each other.
(e) Source and observer moving away from each other.
(f) Source moving away from observer at 10 km/s; observer moving towards source at 5 km/s.
(g) Source moving perpendicular to line of sight. Observer moving towards source.
(h) Source moving and observer moving perpendicular to line of sight but in opposite directions.

5.8. Why must the center of mass of two stars be along the line joining the two stars?

5.9. Why is it not possible for the orbits of the two stars in a binary system to have different periods?

5.10. Is the Earth–Sun center of mass closer to the center of the Earth or the center of the Sun?

*5.11. Suppose that a binary system is moving under the external influence of other stars (in a cluster for example) and we observe the center of mass to be accelerating as a result. Can we still apply the analyses in this chapter to such a system?

5.12. Why do we say that for the Earth–Sun system, the sum of the masses is approximately equal to the mass of the Sun?

5.13. In a binary system, the gravitational force that star 1 exerts on star 2 must produce an acceleration. How is that acceleration manifested?

5.14. Discuss how the Sun is a 'typical' main sequence star.

5.15. How would you measure the mass of the Earth?

5.16. Use a calendar to find out how much longer it takes to go from the first day of Spring to the first day of Autumn than from the first day of Autumn to the first day of Spring (in the northern hemisphere).

5.17. What is the relationship between Kepler's second law and angular momentum?

*5.18. As an object approaches periastron in an elliptical orbit, it is moving faster and faster. This means that there must be a force in the direction of motion (in addition to the force that is perpendicular to the direction of motion, which is responsible for the curving of the path). What is the source of the force that makes the object go faster? Use a diagram, showing the forces and components, to illustrate your answer.

5.19. At apastron the objects in elliptical orbits are moving slower than at periastron, so their kinetic energy must be lower. What happens to the 'missing' energy?

5.20. How does the range of masses for main sequence stars compare with the range of temperatures and the range of luminosities?

5.21. What do we mean when we talk about a selection effect?

5.22. If high mass stars are more massive than low mass stars, how can most of the mass in our Galaxy be in the form of low mass stars? If most of the mass is in low mass stars, how can most of the luminosity come from high mass stars?

5.23. When we study stellar structure, in Chapter 9, we will see that once we know the mass of a main sequence star all of its other properties (size, temperature, luminosity) are determined. How does that show up in the observational results of this chapter?

5.24. Why is it difficult to measure stellar sizes?

Problems

5.1. A police radar, operating at a frequency of 5 GHz, detects a Doppler shift of 2.5 MHz as you approach. How fast are you going?

5.2. (a) A source is moving away from an observer with a speed of 10 km/s, along a path that makes a 30° angle with the line of sight. At what wavelength will the Hα line be observed? (b) If the observer is also moving directly away from the source at a speed of 20 km/s, at what wavelength will the Hβ line be observed?

5.3. Some source is moving away from an observer along a line making a 30° angle with the line of sight at a speed of 80 km/s. The observer is moving towards the source along a line making a 20° angle with the line of sight, at a speed of 10 km/s. (a) What is the radial velocity? (b) By how much is the Hα line shifted? (c) How would the answer be changed if the observer were moving in the opposite direction?

5.4. We observe a source with a radial velocity of −25 km/s. By how much is the separation between the Hα and Hβ lines changed?

5.5. If the Hα line in a source is shifted by +0.01 nm, by how much is the Hβ line shifted?

5.6. If the Hα line in a source is shifted by −0.10 nm, what is the radial velocity of the source?

5.7. You are standing a distance d from a railroad track. A train comes past you at a constant speed v, passing you at time $t = 0$. (a) What is the radial velocity of the train as a function of time? (b) Draw a graph of your result for both positive and negative times.

5.8. By how much can the Hα line in some object be shifted as a result of the Earth's motion around the Sun?

5.9. Show that, if we had taken the force on the second star in equation (5.13), we would still have obtained the result in equation (5.20).

5.10. In deriving the law of gravitation, Newton actually started with Kepler's third law as being observationally given and worked backwards (from the derivation in this chapter). Show how that derivation would be done.

5.11. For a binary system with stars of equal masses m, in circular orbits, with a total separation r, what is the orbital speed v?

5.12. For a binary system with stars of masses 5 and 10 $M_\odot$, in circular orbits with a period of 3 yr, what is the total energy of the system?

5.13. For a binary system with stars of masses m_1 and m_2, in circular orbits, with a total separation r, find an expression for the ratios of the *kinetic* energies of the two stars.

5.14. Let $M = m_1 + m_2$, and $x = m_1/m_2$. (a) Find expressions for m_1 and m_2 in terms of M and x. (b) What is the significance of your result?

5.15. We observe a binary system in which the two stars are 1 and 2 arc sec, respectively, from the center of mass. The system is 10 pc from us. The period is 33 yr. What are the masses of the two stars, assuming that $i = 90°$?

5.16. Suppose we can measure the positions of stars to 0.01 arc sec. How far away could we detect an astrometric binary where the separation is 100 AU?

5.17. A star in a circular orbit has a speed of 30 km/s. The period is 10 yr. The star is 2 arc sec from the center of mass. How far away is this star from us?

5.18. Derive a form of equation (5.23) that relates mass, in solar masses, period, in years, and velocity, in kilometers per second.

5.19. The Hα lines from two stars in a binary system are observed to have Doppler shifts of 0.022 and 0.044 nm, respectively. The period of the system is 20 yr. What are the masses of the two stars (a) if $i = 90°$ (b) if $i = 30°$?

5.20. An astrometric binary is 10 pc from the Sun. The visible star orbits 2 arc sec from the center of mass with a period of 30 yr. What is the mass of the unseen companion?

5.21. An eclipsing binary system has a period of days. One star has a Doppler shift of 100 km/s. What is the mass of the companion?

5.22. Derive equations (5.40a) and (5.40b).

*5.23. A star moves in an elliptical orbit of eccentricity e. The plane of the orbit makes an angle i with the plane of the sky. The orbit is oriented so that the line joining the foci is perpendicular to the line formed by the intersection of the plane of the orbit and the plane of the sky. Show that the projected orbit is an ellipse.

5.24. What is the luminosity (in solar luminosities) of (a) a 0.5 $M_\odot$ and (b) a 5.0 $M_\odot$ star?

5.25. For an elliptical orbit, calculate the angular momentum L in terms of G, m_1, m_2, a and e.

Computer problems

5.1. Plot ellipses with the same major axes, but with eccentricities from 0.1 to 1.0 in steps of 0.1.

5.2. Consider a binary system with stars of masses 5 and 10 $M_\odot$ in elliptical orbits, with eccentricity $e = 0.8$. The period is 8 yr. Assume that the more massive star is at periastron at angle $\theta = 0$. (a) Plot the speed of the more massive star as a function of position in the orbit, θ. (b) If this system is viewed from along the major axis, such that the more massive star is closer to the observer when it is at periastron, plot the radial velocity v_r as a function of θ. *(c) Plot v_r as a function of time t, taking $t = 0$ when $\theta = 0$.

5.3. For the system in the previous problem, draw the orbit and draw an arrow showing the force on the more massive star at positions every 45° around the orbit.

5.4. Suppose that many binary star systems, with randomly distributed inclinations, are observed. (a) Draw a graph showing the relative probabilities of finding values of $\sin^3 i$ in various small ranges from zero to one. (b) The average value of some function $f(x)$ over the interval 0 to L is

$$\langle f(x)\rangle = \left(\frac{1}{L}\right)\int_0^L f(x)\,dx$$

What is the average value of $\sin^3 i$ over the angle range 0 to $\pi/2$ radians? (c) What does this tell you about the lower limits on masses and the actual masses of binary systems?

5.5. For stars in the middle range of each spectral type (05, B5, etc.), calculate the average density, and express it as a ratio to that for the Sun.

5.6. For stars in the middle range of each spectral type (05, B5, etc.), calculate the luminosity, and express it as a ratio to that for the Sun.